

NGS Result Report

Report ID: stage101-fully_populated-report	Input allele: 1
Genome build: GRCh38	Report state: Confirmed
Generated: 2026-08-11	Template: professor-report-v1

Clinical Features:

Accepted features: Neurologic abnormality (HP:0000707); Seizure (HP:0001250). Phenotype concordance: partially supported. One accepted synthetic feature has source-supported gene context; the remaining feature is not established by the retained evidence.

Method:

Allele-level evidence synthesis for an already filtered variant; this application did not perform sequencing or genome-wide prioritization. Call quality — QUAL: 99; FILTER: PASS; state: passed.

Variant and source classification context:

GENE1:c.101C>T p.(Arg34Trp)

Heterozygous

System classification: Not independently determined

Source: No independent application classification

Brief Interpretation(s):

The synthetic GENE1 c.101C>T allele has rare exact-allele population evidence, uncertain-significance ClinVar context, automated GeneBe context, and three source-attributed computational predictions. Phenotype support is partial rather than complete. The evidence does not support an independent application classification, and the report preserves source roles, missingness, and the fallback population record for qualified human review.

Decision-support output for educational and research use; qualified human review is required.

Main Finding(s) in Detail:

Gene & transcript	Variant	Population frequency			Associated disease & ID	Zygosity
		gnomAD gnomAD r4	1000 Genomes Phase 3	Ensembl Variation GRCh38 variation (fallback)		
GENE1 NM_000001.1	c.101C>T p.(Arg34Trp)	1.230e-05	No exact record	2.000e-05	Synthetic neurodevelopmental condition (MONDO:0000001)	Heterozygous
SIFT	PolyPhen-2	CADD		Stable variant ID	Classification context	Inheritance
deleterious score 0.01	probably damaging score 0.97	elevated score score 24.1		fixture ID: SYNTHETIC- VARIANT- 001	NCBI ClinVar: Uncertain significance (VUS) (criteria provided, single submitter) GeneBe: Automated uncertain significance (automated annotation only)	autosomal dominant

Population values are exact-allele observations from the named source. Computational predictions are source-attributed and are not independent ACMG classifications. Missingness is written explicitly.

Variant interpretation:

GENE1:c.101C>T p.(Arg34Trp)

This synthetic GENE1 allele is rendered to exercise the complete report pathway. The retained gene-disease record provides contextual support for a synthetic neurodevelopmental condition with autosomal dominant inheritance, but it does not establish that this allele is causal. The allele is rare in the primary population dataset, was not identified in the retained 1000 Genomes exact query, and has a source-preserving fallback population observation.

NCBI ClinVar reports an uncertain-significance classification with single-submitter review context. GeneBe supplies an automated uncertain-significance annotation; this remains attributed automated evidence and is not adopted as an application classification. SIFT, PolyPhen-2, and CADD results provide concordant computational concern, but computational predictions do not independently establish pathogenicity.

The accepted phenotype is partially supported: one synthetic neurologic feature overlaps retained gene

context, while the remaining feature is not established by the available evidence. The literature entries describe general interpretation standards and are not allele-specific proof. No material classification conflict is resolved by the application. Overall, the evidence remains insufficient for an independent ACMG/AMP conclusion and requires qualified human review.

Variant(s) classification:

The retained sources provide uncertain-significance context without an independently adjudicated application classification. Reviewer-confirmed classification: Not available. ClinVar classification: Uncertain significance (VUS). Automated classification context: GeneBe automated uncertain significance. Conflict status: none. Sources: NCBI ClinVar, GeneBe automated annotation. The application has not independently adjudicated an ACMG/AMP classification.

Method:

The application validated the supplied allele identity, collected bounded source-attributed evidence, preserved provider availability and fallback provenance, and rendered the reviewer-approved interpretation without a new model request.

Comments / scope:

All allele and phenotype values in this golden report are synthetic and reserved for layout validation. One population observation was supplied by the explicitly identified fallback source. The CSpec record is context only; no specification criteria were executed. This allele-level report supports qualified human review and does not provide a diagnosis, treatment recommendation, or testing directive.

References:

1. Standards and guidelines for the interpretation of sequence variants. [PMID: 25741868](#)
2. ACMG and AMP sequence variant interpretation guidance. [DOI: 10.1038/gim.2015.30](#)
3. Sequence variant interpretation standards. [PMCID: PMC4544753](#)

Data Sources:

Source	Capability	Status	Record or dataset	Method	Link
Ensembl VEP	variant consequence	Available	SYNTHETIC-VARIANT-001	exact allele query	Open record
NCBI ClinVar	clinical classification	Available	synthetic-clinvar-context	exact allele query	Open record
GeneBe	automated classification context	Available	2-100001-C-T	exact allele query	Open record
ClinGen / GenCC	gene disease context	Available	MONDO:0000001	gene context query	Open record
ClinGen CSpec	specification context	Available	synthetic-context-only	released metadata lookup	Open record
gnomAD	population frequency	Available	gnomAD r4	exact allele query	Not available
Ensembl Variation	population frequency	Available (fallback)	SYNTHETIC-VARIANT-001	exact overlap query	Open record
Europe PMC	literature retrieval	Available	MED:25741868	identifier lookup	Open record